



Shilpa

**Inclusive Mobile Learning App For Visually,
Hearing, Physically And Cognitively
Impaired Students (Grade 3-5)**

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Introduction

- Designed for Grade 3–5 students with disabilities
- Supports visual, hearing, physical, and cognitive challenges
- Provides special learning tools:
- Voice commands, subtitles, large buttons, simple visuals
- Sinhala Sign Language support
- Includes fun quizzes and games to improve engagement
- Auto-generated quizzes to measure learning outcomes
- Parents and teachers can track student progress
- Ensures equal learning opportunities for every child





Problems



- Digital learning tools used in Sri Lanka are not designed for inclusive primary education.

- Most applications:

Address only a single disability

Lack Sinhala-first support

Are not aligned with the Grade 3–5 national curriculum

- Students with disabilities rely heavily on external assistance to complete basic learning activities.
- Lack of fun, interactive learning tools





Impact / Gravity

- Low engagement and motivation in foundation grades negatively affect learning outcomes
- Difficulty in personalizing learning and tracking student progress for teachers and parents
- The education gap for differently-abled learners continues to widen
- Provides equal learning opportunities for visual, hearing, physical, and cognitive disabilities
- Interactive and adaptive learning methods improve student engagement
- Progress tracking and feedback enhance teacher and parent involvement





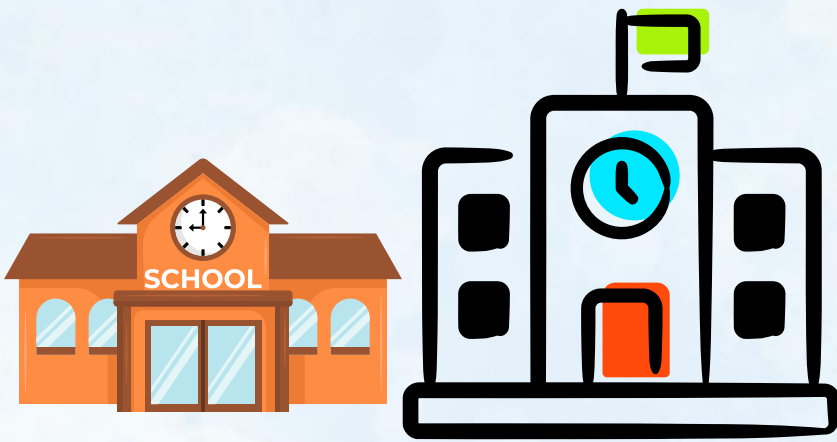
Target Users & Beneficiaries



- **Primary Users:** Grade 3–5 students with visual, hearing, physical, and cognitive disabilities



- **Secondary Users:** Teachers and parents for lesson personalization and progress tracking



- **Institutions / Society:** Schools, special education centers, and the Sri Lankan education system



- **Technology Need:** Inclusive, Sinhala-first, adaptive mobile learning platform



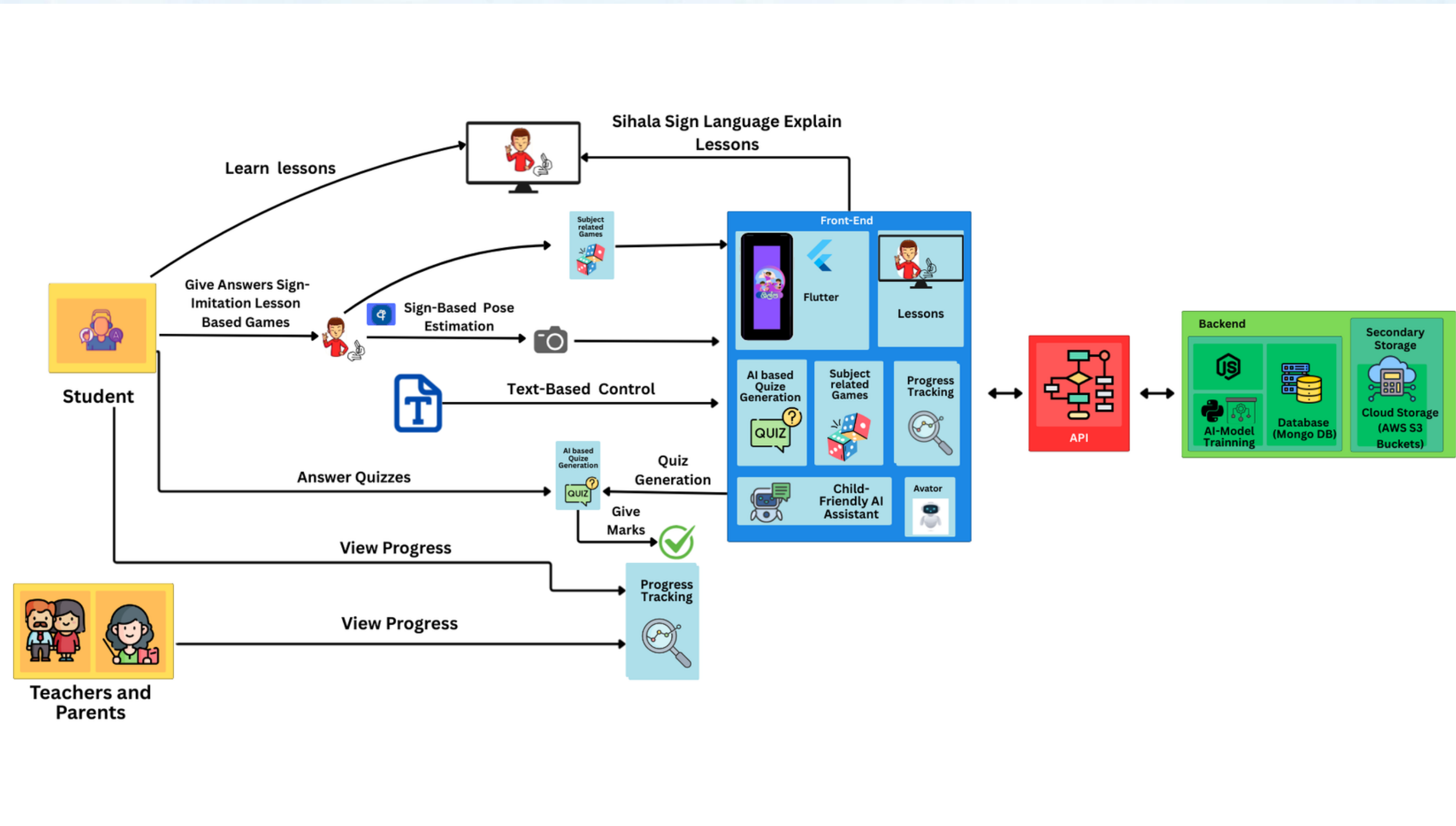
Solutions

- Inclusive mobile learning application for Grade 3–5 students with disabilities
- Single, unified platform supporting visual, hearing, physical, and cognitive impairments
- Sinhala-first, curriculum-aligned primary education content
- Adaptive and accessible UI with disability-specific interactions
- Integrated assistive technologies including TTS, STT, subtitles, sign-based learning, voice, gesture, and gaze control
- Personalized interactive quizzes to improve learning outcomes
- Progress tracking dashboards for teachers and parents
- Modular and scalable architecture enabling future enhancements and wider adoption





Overall System Diagram





Hearing Impairment Support



BSc (Hons) in Information Technology
Specializing in Information Technology





Hearing Impairment Support



Problems

- Teachers use Sinhala Sign Language with speech in classrooms, but digital learning tools do not support this dual-mode teaching well.
- Most existing applications use ASL/BSL, not Sinhala Sign Language (SLSL)
- Digital tools are mostly video-based, with no active sign practice
- No real-time feedback or gesture scoring for sign performance
- Limited support for independent learning at home



Solutions

- Lessons delivered using Sinhala Sign Language with teacher speech together
- Lesson-based sign imitation games aligned with Grade 3–5 curriculum
- Camera-based hand gesture recognition for real-time feedback
- No external devices required (mobile camera only)
- Game-based learning to improve engagement and confidence



User Requirements Addressed

- Supports fully deaf and partially deaf students
- Visual, text, sign, and speech-supported learning
- Simple, child-friendly UI with Sinhala instructions
- Interactive practice to improve sign accuracy



Hearing Impairment Support



Design Excellence / Novel Contribution

- Lesson-based sign imitation games (**NOVELTY**)
- Signs are practiced inside lessons, not as separate videos
- Students actively perform signs, not just watch demonstrations
- Gesture scoring (0–100) instead of only correct / incorrect results
- Optimized for smooth mobile performance



AI / ML Technologies Used

- MediaPipe Hands for real-time hand landmark detection
- Landmark normalization to handle hand size and position differences
- **Logistic Regression**
Used for number sign recognition
Lightweight and fast for mobile use
- **Bidirectional LSTM**
Used for sequence-based sign recognition
Learns sign movement over 30-frame sequences.
- On-device inference for real-time feedback



Current Progress (PP1 Stage)

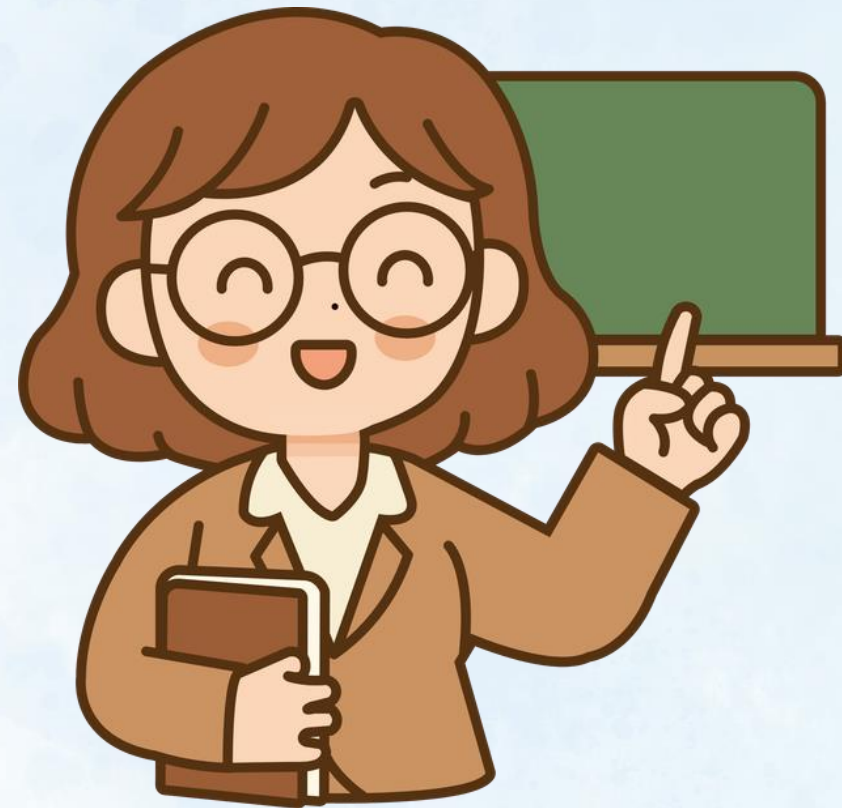
- User registration & login
- Subject & lesson selection
- Sign imitation games using camera
- Quiz generation



Hearing Impairment Support

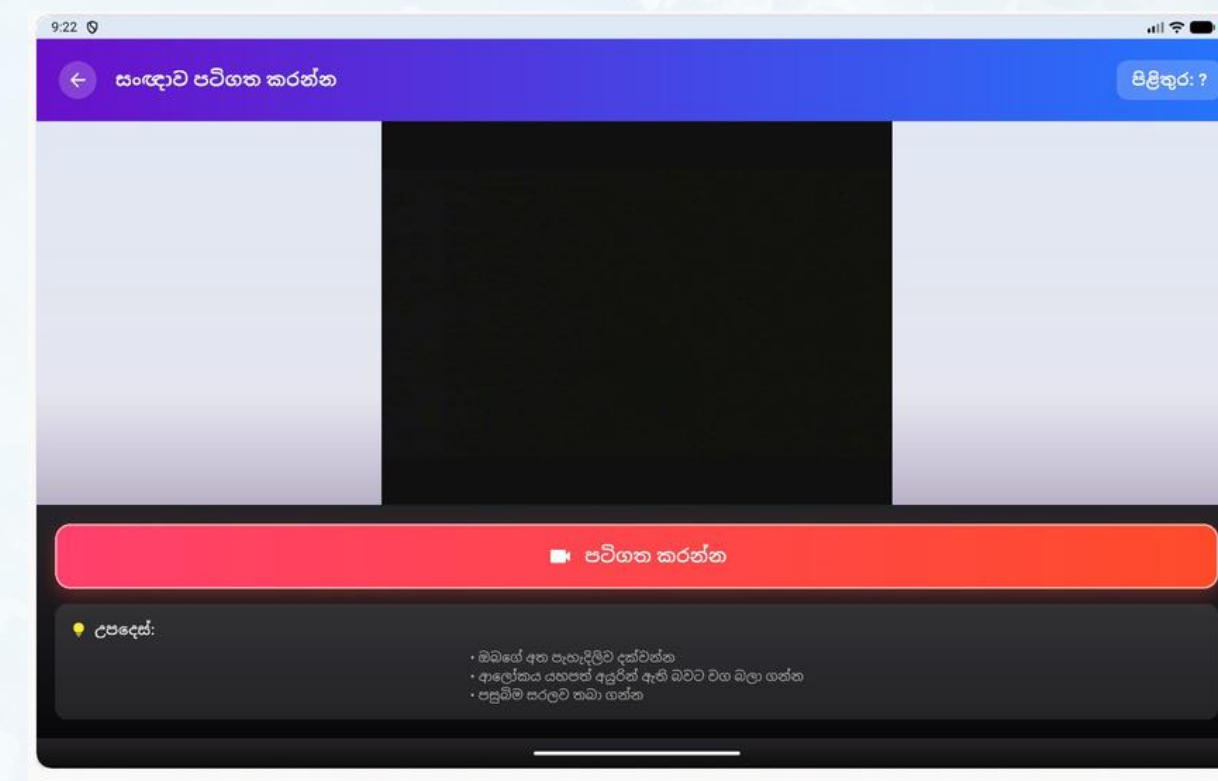
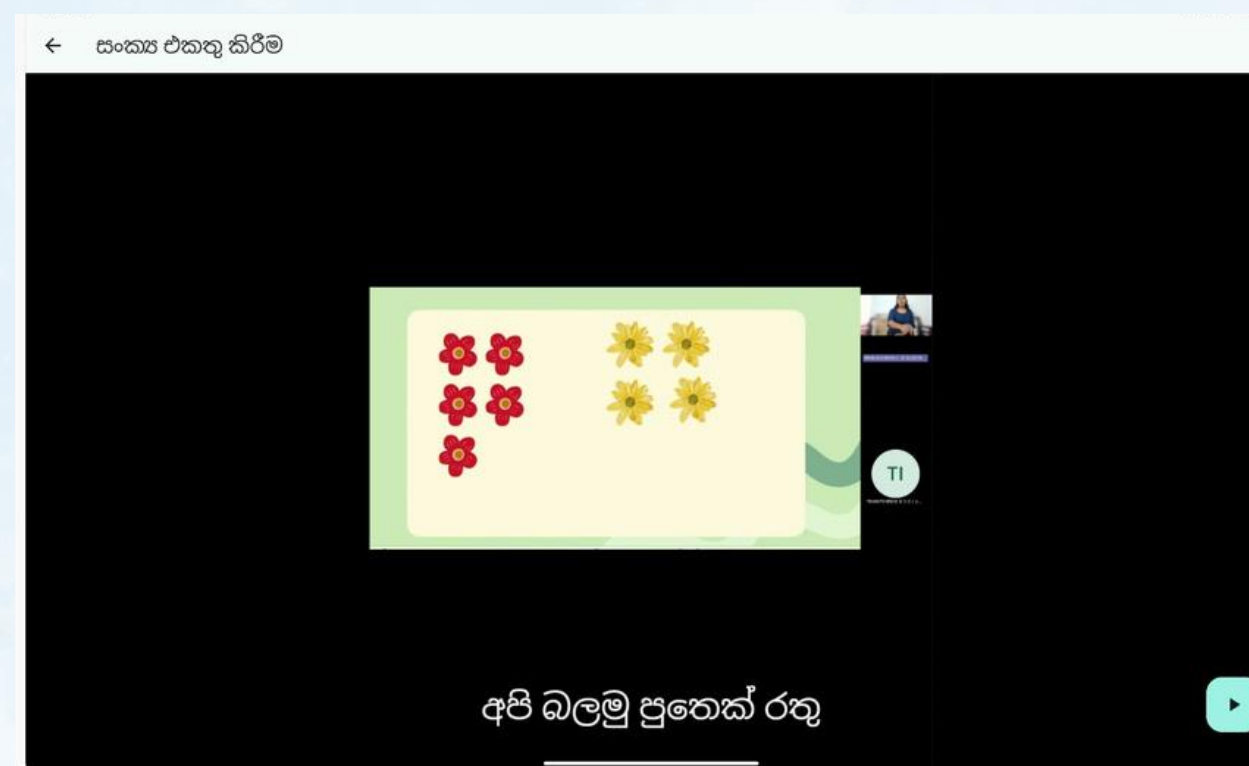
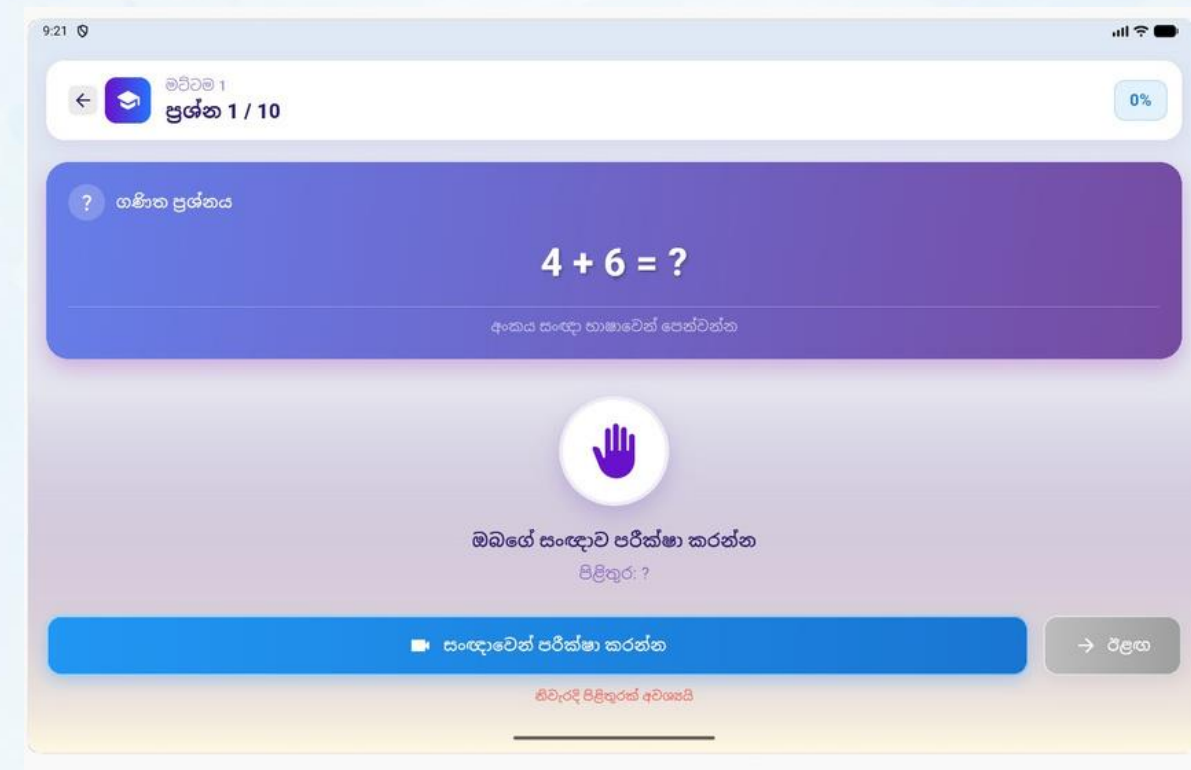
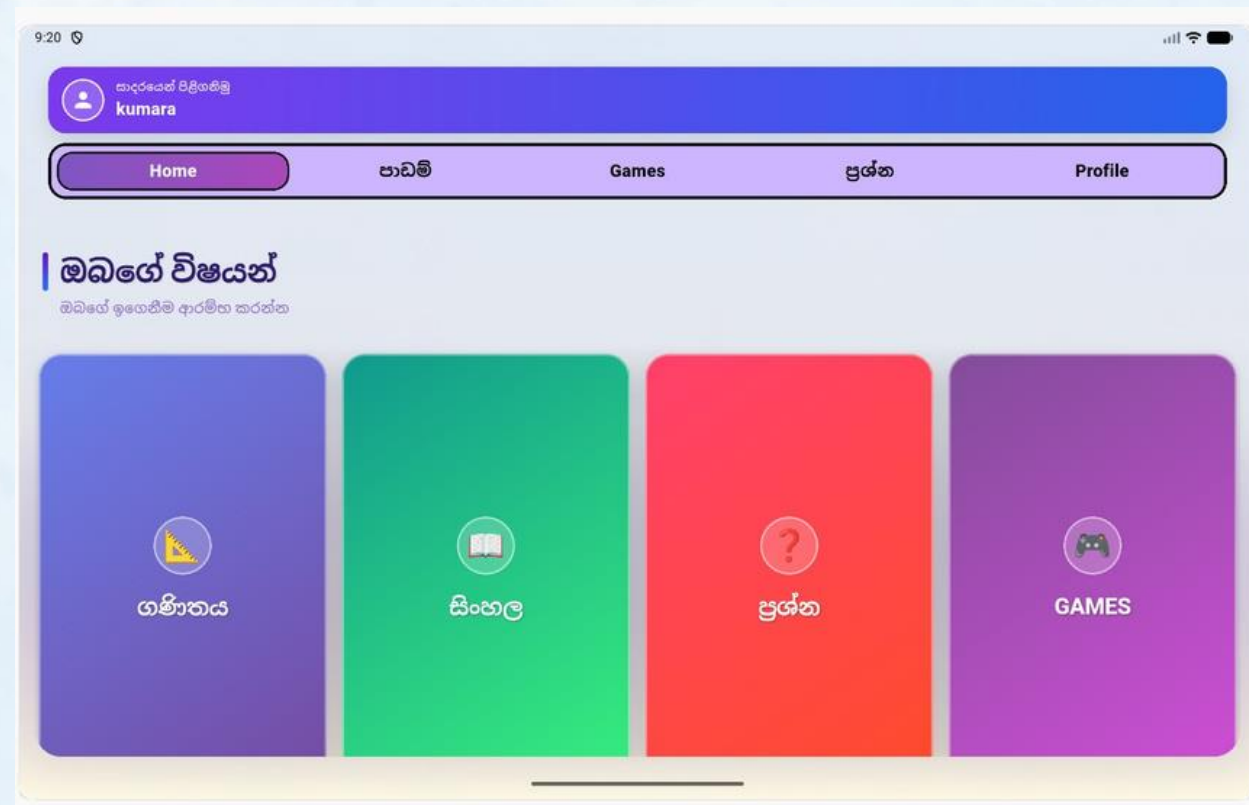
User Feedbacks

- Teachers stated that lesson delivery using both spoken explanations and SLSL (Sri Lankan Sign Language) is effective for students.
- Teachers appreciated the use of image-based lessons, as visuals help students understand concepts more clearly.
- Teachers suggested using quizzes with typed answers, since some students are less motivated to write by hand.
- Teachers observed that students are more engaged and motivated when learning through technology.





Hearing Impairment Support





Visual Impairment Support



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Visual Impairment Support



Problems

- Visual lesson content is inaccessible for fully blind students.
- Poor UI design (small text, low contrast) affects partially blind learners.
- Single systems do not support different vision levels effectively.
- Visual-based assessments are difficult for blind students.
- Most apps rely only on basic text-to-speech without curriculum alignment.
- 3D spatial audio is rarely used to support concept understanding.



Solutions

- Voice-guided lessons and navigation for fully and partially blind students.
- 3D spatial audio to simulate real environments and support understanding.
- Accessible UI with large text, clear icons, and user-friendly color themes.
- Swipe-based answering instead of visual buttons.
- Curriculum-aligned content for Grades 3–5.
- Optional automatic Braille document generation for school-based printing.



User Requirements Addressed

- Supports fully blind and partially blind students
- Voice-guided learning for lesson delivery and navigation
- High-contrast, child-friendly UI with large text for low-vision users
- 3D spatial audio to improve understanding of lesson concepts
- Simple swipe-based interaction for answering questions
- Curriculum-aligned content for Grade 3–5 students
- Optional Braille lesson and question documents for school-based printing



Visual Impairment Support



Design Excellence / Novel Contribution

- Voice-guided learning for lessons and navigation.
- 3D spatial audio for improved concept understanding.
- Single system for fully and partially blind students.
- Accessible, child-friendly UI with high contrast and large visuals.
- Automatic Braille document generation for school use. **(NOVELTY)**
- Vibration and sound feedback for interactions and game completion.
- Curriculum-aligned design for Grade 3-5 learners.



AI / ML Technologies Used

- Sinhala Text-to-Speech (TTS)
- Speech-to-Text (STT)
- Voice-guided interaction
- Vibration-based feedback
- FFmpeg



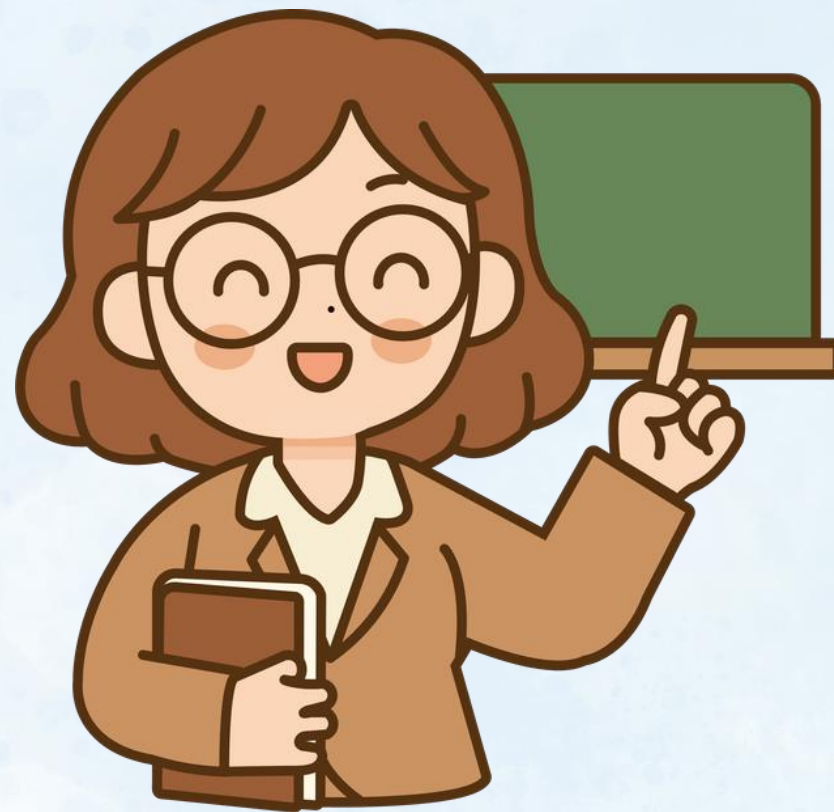
Current Progress (PP1 Stage)

- Voice-based navigation
- Text-to-Speech (TTS) for all learning content
- Voice-guided quizzes
- Braille-ready document generation for quizzes
- Vibration feedback for actions



Visual Impairment Support

User Feedbacks

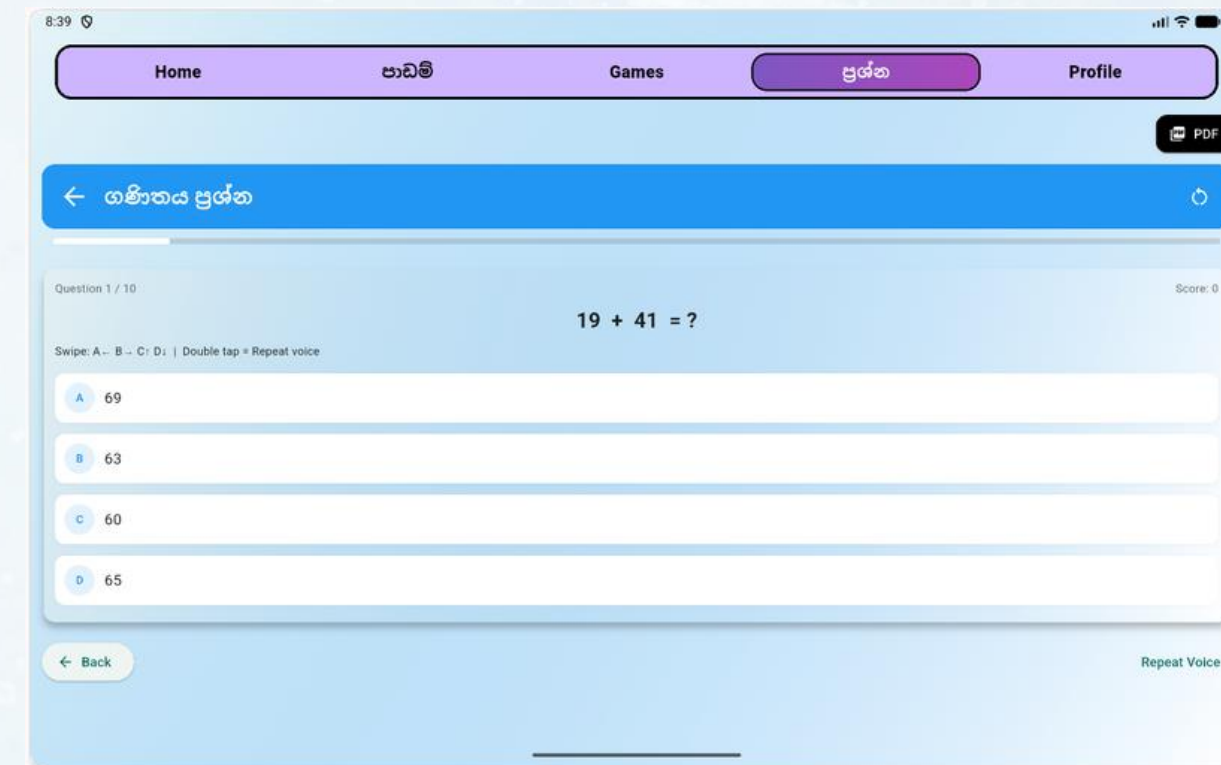
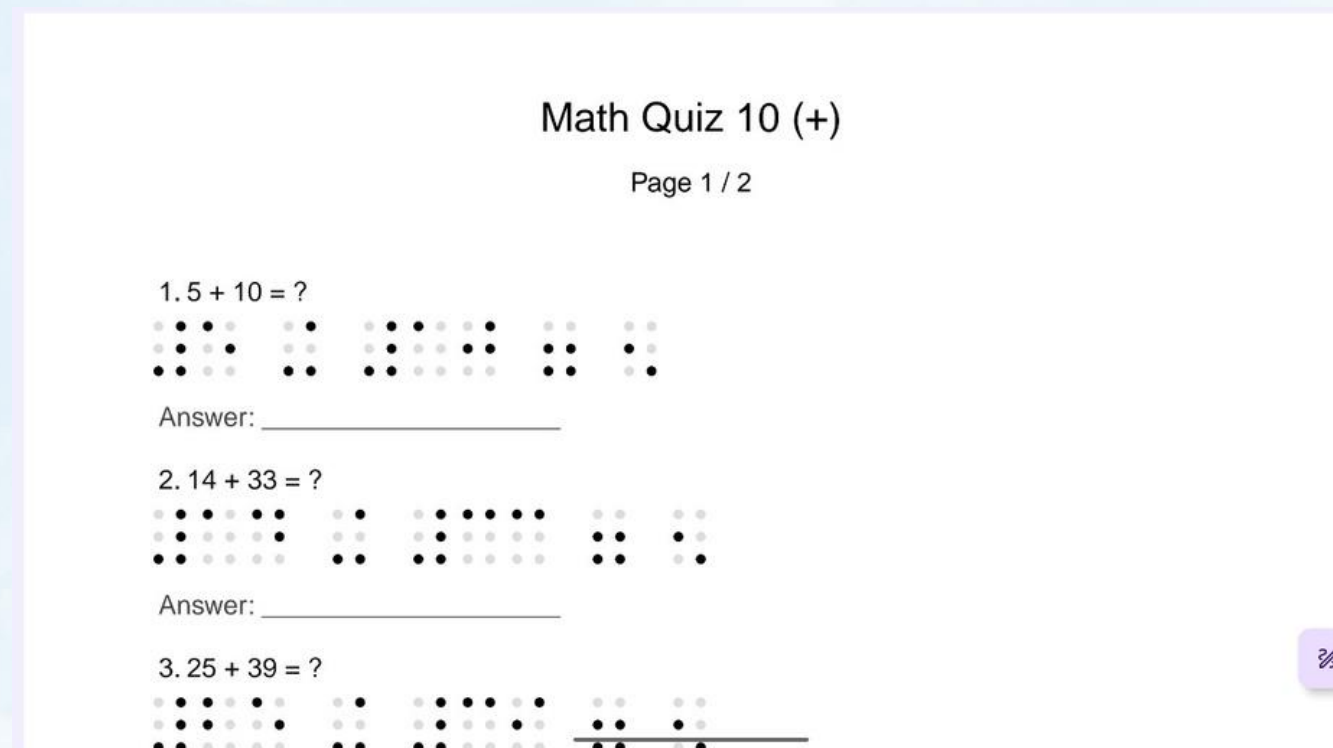
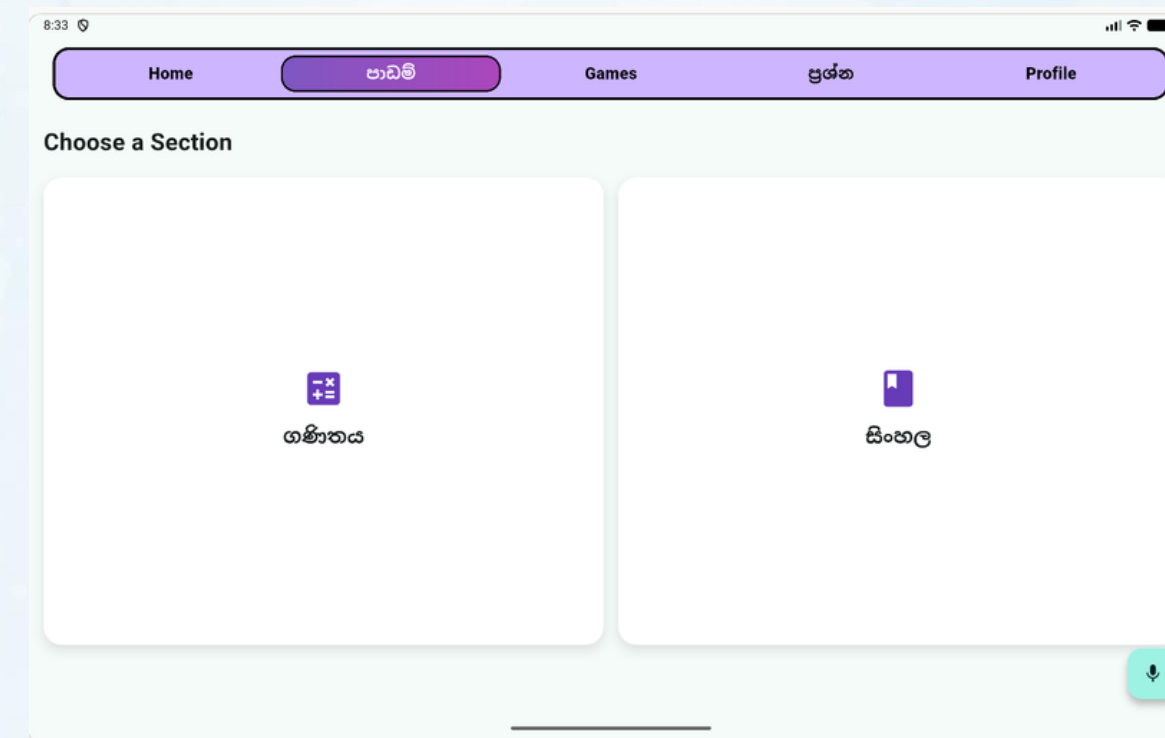
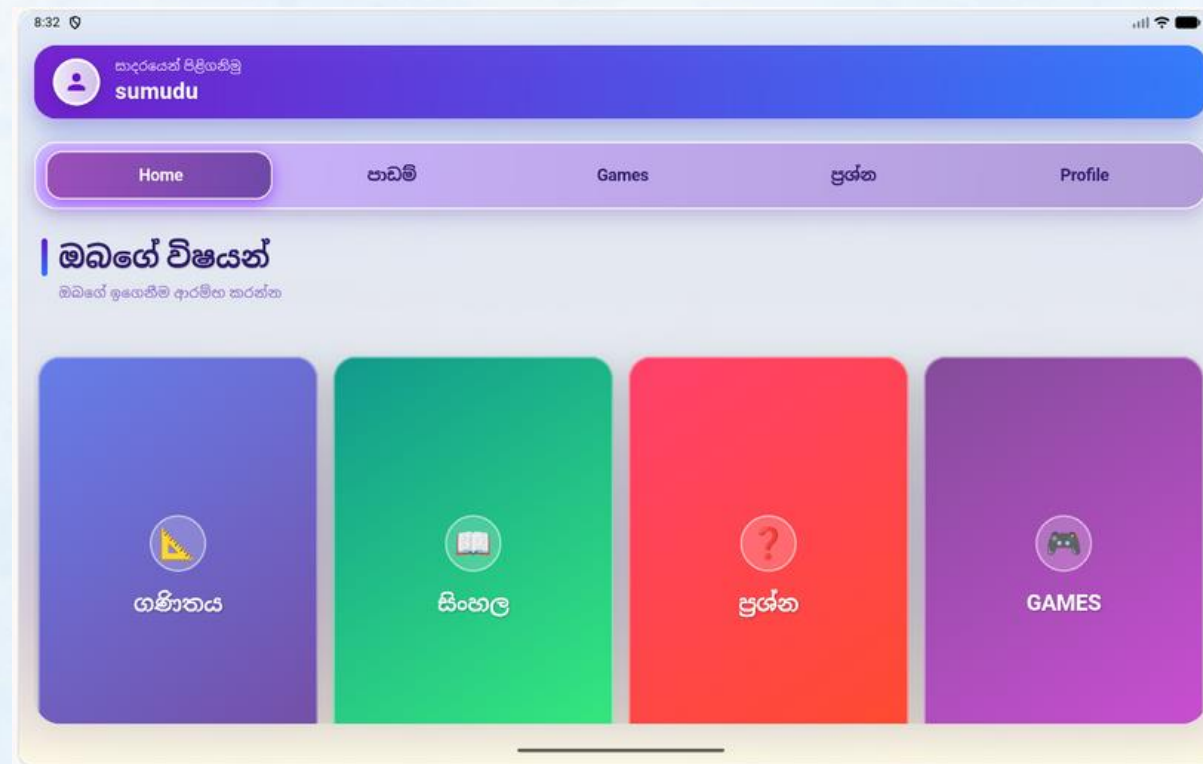


- Teachers advised that both partially sighted and fully blind students should use the same Braille-based learning method, as examinations are conducted using Braille.
- Teachers recommended story-based lessons to help students understand concepts more effectively.
- Teachers emphasized that lessons should be delivered with slow and clear explanations, since students cannot rely on visual cues.
- Teachers suggested using 3D or spatial audio, so students can better experience and understand the learning environment.





Visual Impairment Support





Physical Impairment Support



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Physical Impairment Support



Problems

- Not providing Safety Features.
- Lack of child friendly environment with simple UIs.
- Lack of multiple options to control (Especially for Physically Disabled People)



Solutions

- Providing multiple handling options
- Customizable Buttons and Fonts
- Providing Accidental Touch protection options and Emergency **Stop** command feature

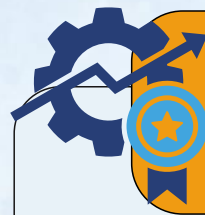


User Requirements

- Support for Physically Impaired students (Students who are paralyzed / partially physically disabled)
- Able to use Dwell time, Head/ Eyes Gesture Navigation, and Voice command features
- Providing Large and pressable buttons



Physical Impairment Support



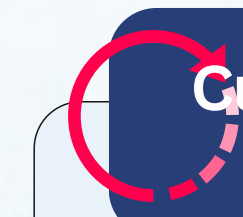
Design Excellence / Novel Contribution

- Providing multiple options to handle the app (dwell, Eye gaze, and Voice), considering movement difficulties. **(Novelty)**
- Accidental Touch detection option and **“Cancel”** command to terminate the action.
- Large and Pressable buttons for easy engagement with the app



AI / ML Technologies Used

- Google ML Kit for face detection and eye/head gestures
- STT (Speech-to-Text) Flutter plugin for voice navigation.
- Gesture detection widget using standard Flutter libraries



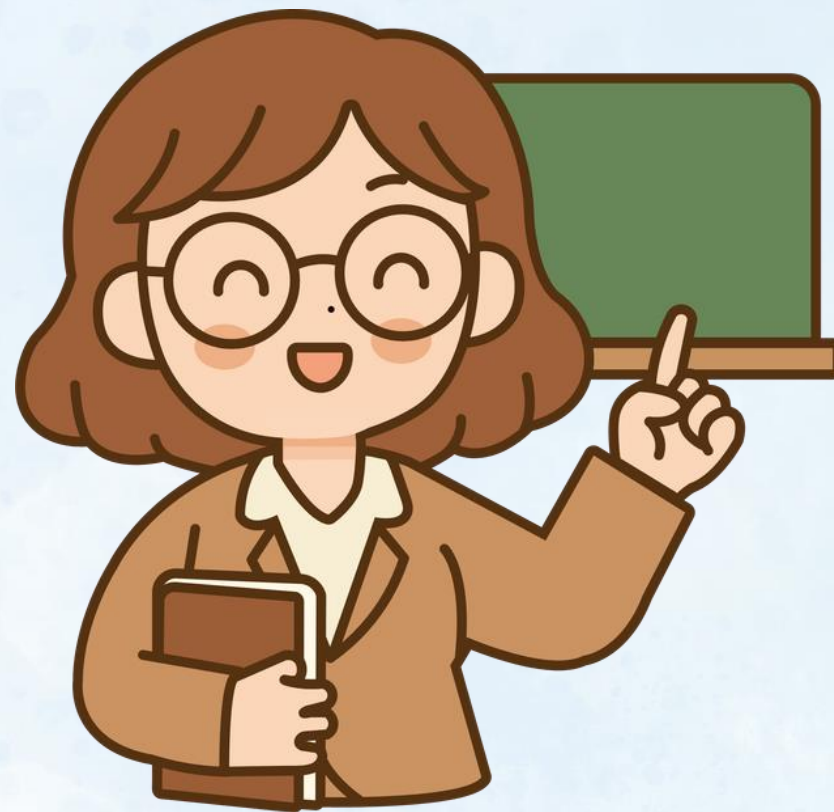
Current Progress (PP1 Stage)

- Implemented the dwell touch feature with dwell detection
- Fixed smooth eye/head gaze tracking feature (with clarification and showing accuracy)
- Done Voice-based navigation with several simple commands that can navigate through the main pages



Physical Impairment Support

User Feedbacks

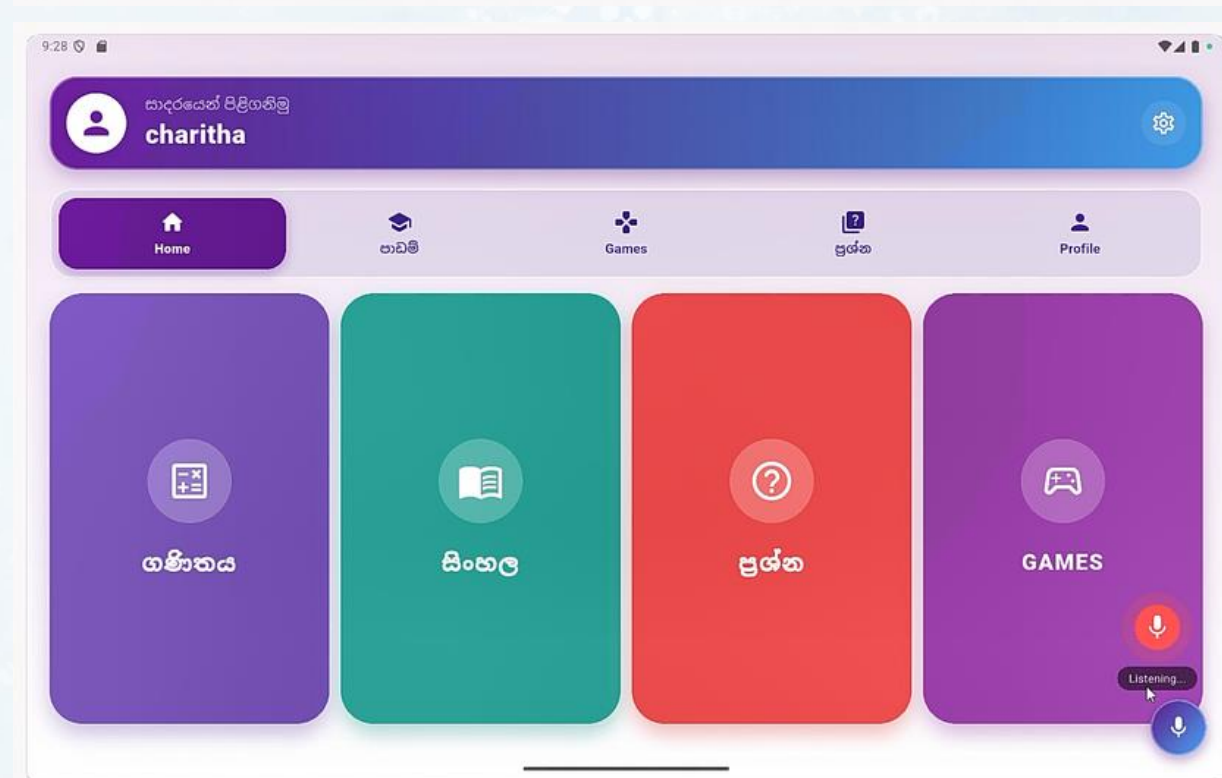
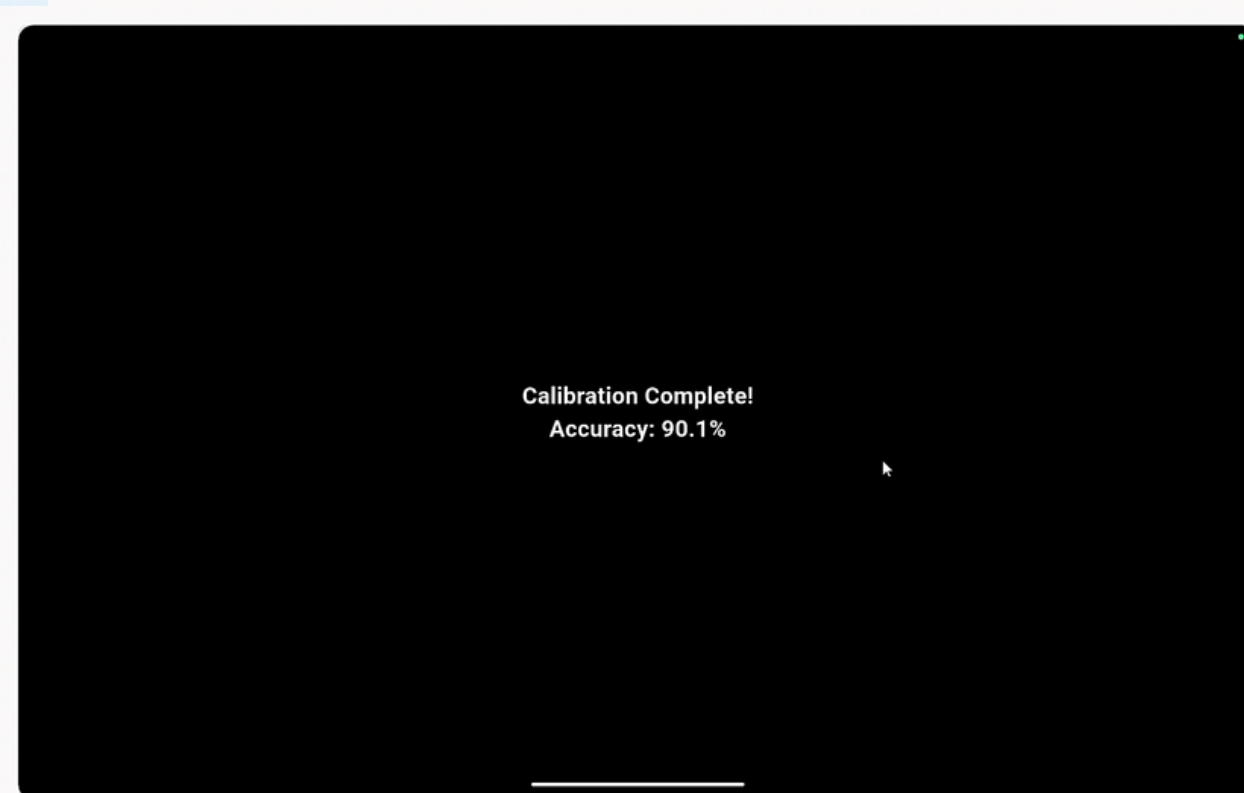
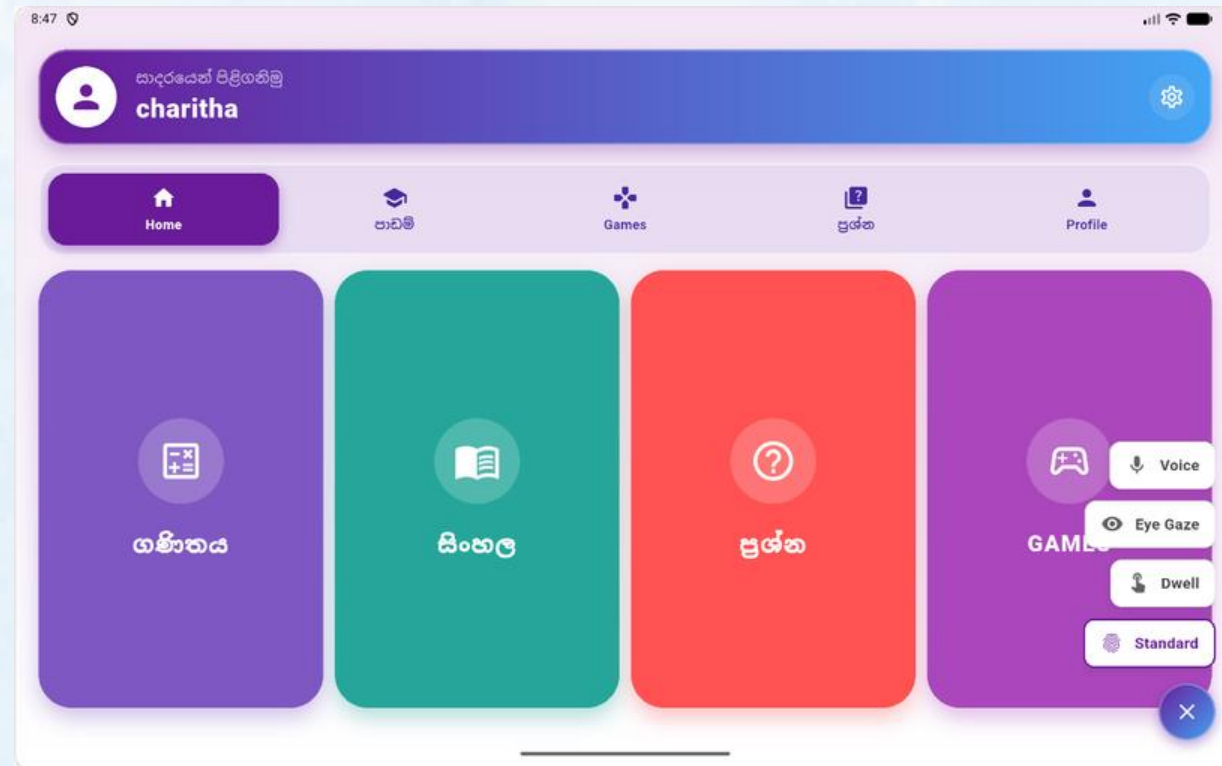


- Make the buttons large and make them pressable
- Make the eye gaze control less engaged and smoother
- Use simple commands, either English or Sinhala doesn't matter





Physical Impairment Support





Cognitive & Learning Difficulties Support



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Cognitive & Learning Difficulties Support



Problems

- Students with cognitive and learning difficulties often struggle with focus, memory, and understanding in regular classroom lessons.
- One-size-fits-all teaching does not adapt to different learning speeds or IQ levels.
- There are very few Sinhala-first digital learning platforms that offer personalized, supportive, and adaptive learning experiences.



Solutions

- Learning activities automatically adjust to each student's ability and performance.
- Short, game-based lessons make learning easier and less stressful.
- Friendly animated avatars give encouragement, hints, and instant feedback.
- Lessons are delivered mainly in Sinhala, helping students feel more comfortable and confident while learning.



User Requirements Addressed

- Supports slow, step-by-step learning with enough practice time.
- Uses visuals, sounds, and interactions to make learning easier and less tiring.
- Lets teachers and parents easily track student progress.
- Designed to be child-friendly, simple, and easy to use for learners with special needs.



Cognitive & Learning Difficulties Support



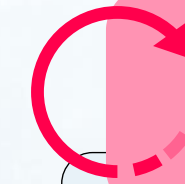
Design Excellence / Novel Contribution

- Personalizes learning based on IQ levels instead of using one fixed lesson style.
- Uses simple activities at the beginning to understand the student's learning level before starting lessons. (NOVELTY)
- Brings gamification, emotional support, and adaptive learning into one app.
- Uses animated avatars as learning companions to guide and motivate students.



AI / ML Technologies Used

- Uses supervised machine learning to group students based on performance.
- A neural network model (TensorFlow / Keras) predicts cognitive performance fairly and accurately.
- The model is optimized with TensorFlow Lite for smooth, on-device use in the Flutter app.



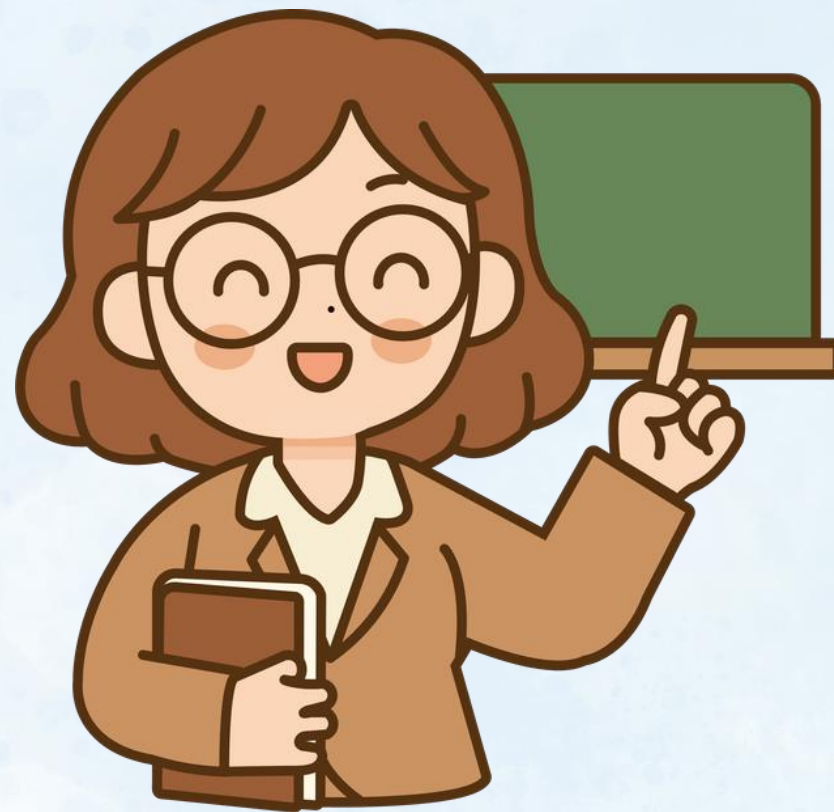
Current Progress (PP1 Stage)

- The main learning flow and IQ-based grouping are designed and tested.
- Early versions of game-based activities and adaptive rules are working.
- Child-friendly and accessible UI designs are 90% completed.
- Basic progress tracking and feedback are already functional.



Cognitive & Learning Difficulties Support

User Feedbacks

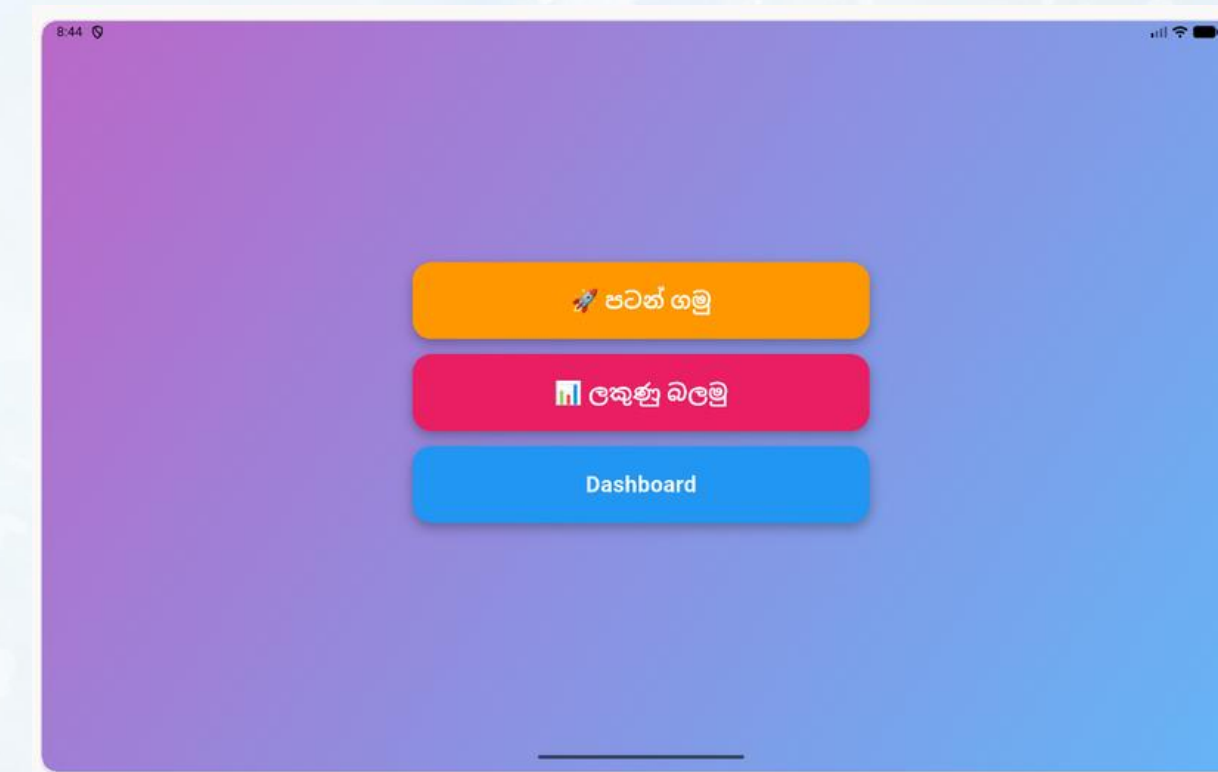
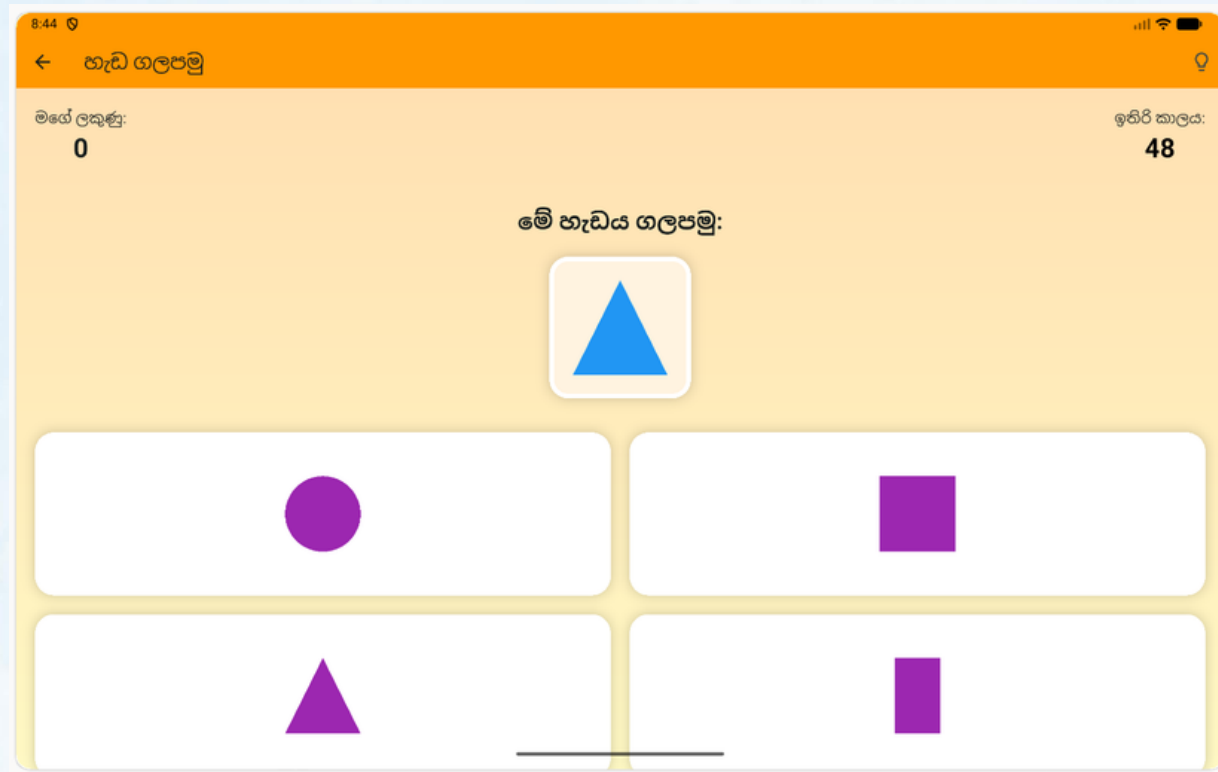
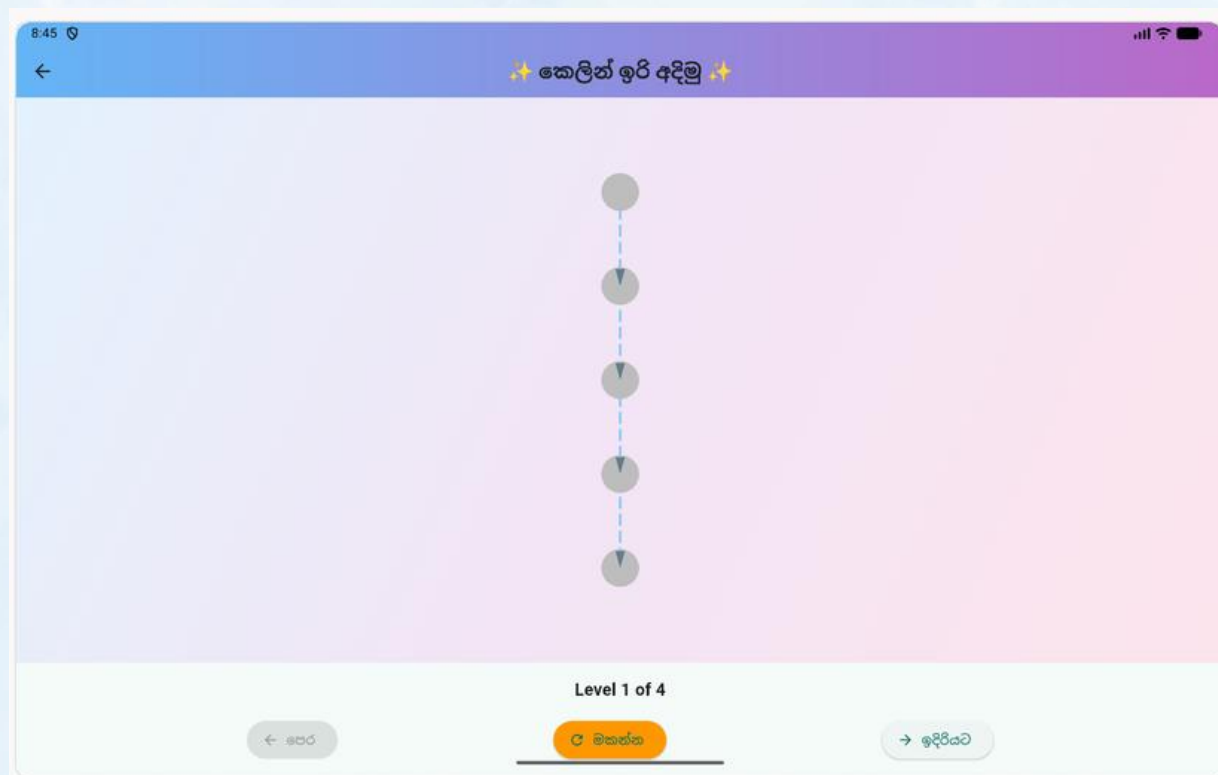


- Teachers stated that simple navigation helps students with cognitive and learning disabilities move through the application without confusion.
- Teachers observed that an easy-to-identify interface allows students to recognize buttons, actions, and content more quickly.
- Teachers appreciated the child-friendly user interface, as it keeps students comfortable, focused, and motivated during learning.
- Teachers emphasized that simple and clear lesson content supports better understanding and reduces cognitive overload for students.





Cognitive & Learning Difficulties Support





Shilpa -Inclusive Mobile Learning App For Visually, Hearing, Physically And Cognitively Impaired Students (Grade 3-5)

Shilpa – Commercialization & Sustainability Customer Persona



Primary customers:

- Parents of Grade 3–5 students with disabilities
- Special education schools and inclusive primary schools

Secondary customers:

- Teachers, special educators, and NGOs
- Government and education-related institutions

Market Size



- A large number of primary school students with disabilities in Sri Lanka
- Growing demand for digital and inclusive education platforms

Potential expansion to:

- Other grades (Grade 1–10)
- Other languages and regions

What is Unique in Shilpa



- Single unified app supporting visual, hearing, physical, and cognitive disabilities
- Sinhala-first, curriculum-aligned primary education content
- Lesson-based sign imitation games (key novelty)
- Combines learning, assessment, and progress tracking in one platform
- Designed specifically for Sri Lankan education context

Cost Recovery / Sustainability Model

- Subscription model for schools and institutions
- Freemium model for parents (basic free, advanced features paid)
- Institutional licensing for special education centers
- Future partnerships with NGOs and government education programs





THANK YOU

